StuTech Science Notes

Oceans and Marine Life

Introductory marine science notes

Learning goals

• Describe major ocean zones.

• Explain how energy moves through a marine food web.

• Identify adaptations and conservation challenges in marine environments.

1. Why oceans matter

Oceans cover about 71% of Earth’s surface. They store and move heat, influence weather and climate, support immense biodiversity, and provide food and livelihoods. Ocean water is salty because dissolved ions enter through rock weathering, rivers, hydrothermal processes and other natural sources.

2. Ocean zones

Zone or region	Description
Intertidal	Shore area covered and uncovered by tides.
Neritic	Shallow water over the continental shelf; often productive.
Open ocean	Waters beyond the continental shelf.
Sunlit zone	Upper ocean where enough light supports photosynthesis.
Deep ocean	Dark, cold, high-pressure environment below the sunlit zone.

3. Marine food webs

Phytoplankton use sunlight to make organic matter and form the base of many marine food webs. Zooplankton eat phytoplankton and are eaten by small fish and other animals. Larger fish, seabirds, marine mammals and decomposers are connected through feeding relationships. A food web is more realistic than a single food chain because organisms often have multiple food sources and predators.

4. Adaptations to ocean life

Fish often use gills to obtain dissolved oxygen. Many marine mammals breathe air with lungs and must surface. Some animals use camouflage, schooling, bioluminescence or specialized body shapes to survive. Organisms in deep water may be adapted to darkness, cold temperatures and high pressure.

5. Ocean challenges and conservation

Marine ecosystems face pressures from plastic pollution, warming waters, ocean acidification, habitat destruction, overfishing and nutrient pollution. Helpful actions include reducing single-use plastics, disposing of waste responsibly, choosing sustainable seafood where possible, protecting coastal habitats and supporting evidence-based conservation.

6. Review questions

1. What role does phytoplankton play in ocean food webs?
2. How do marine mammals differ from fish in breathing?
3. Name two human-caused pressures on marine ecosystems.
4. Why are coastal wetlands and reefs valuable habitats?

Study tip: Read each section, write a short summary in your own words, then answer the review questions without looking at the notes.

Educational resource prepared for the StuTech demo library. Check your course syllabus and teacher guidance for examinable detail.